

Online Internship on Open-Source Processor Architecture (IBM OpenPOWER) and SoC Design

Assignment No. 07: Microwatt Overview

| Instructions to Students

Answer all the questions below based on the Microwatt lecture session. Please upload your complete compiled answers, along with any diagrams, VHDL code snippets, or screenshots referred to, as a single PDF file.

| Assignment Questions

- Q1. What is Microwatt, and which ISA does it implement? Name the organization that develops it and the HDL it is written in.
- Q2. List any four architectural features of the Microwatt ISA (e.g., register count, addressing width, endianness support).
- Q3. Draw/describe the 5-stage pipeline of the Microwatt core and briefly explain the function of each stage.
- Q4. Why does Microwatt not implement branch prediction? What technique does it use instead?
- Q5. Explain the role of the icache and dcache modules in the core. What signals do they use to interface with the Wishbone bus?
- Q6. What is the purpose of the mmu module in Microwatt, and what happens on a TLB miss?
- Q7. List any five SoC-level peripheral components (outside the core) and describe their function in one line each.
- Q8. Explain the role of xics_ics and xics_icp in Microwatt's interrupt handling.
- Q9. Describe the power-up and reset sequence of the Microwatt SoC, from Power-Up to SoC Ready for Full Operation.
- Q10. Name three configurable generic parameters of the Microwatt core entity and explain what each controls.